

AP CALCULUS AB - UNIT 4

Guided Notes 4.4: Related Rates



Name: _____ Class: _____ Date: _____

Learning Objectives and Essential Question

Essential question: How can we identify changing quantities linked by a geometric or physical constraint?

- Identify changing quantities linked by a geometric or physical constraint.
- Differentiate a constraint with respect to time using implicit differentiation.
- Substitute information from a specified instant and solve for an unknown rate.
- Interpret the sign and units of the resulting rate.

Key Knowledge, Vocabulary, and Notation

Related rates

Rates of two or more quantities connected by an equation and changing with a common variable, usually time.

Constraint

An equation relating the changing quantities, such as $x^2 + y^2 = z^2$ or $V = \frac{1}{3}\pi r^2 h$.

Differentiate first

Keep variables symbolic while differentiating; substitute the instant afterward.

Chain Rule

$$\frac{d}{dt}(x^n) = nx^{n-1}x' \text{ because } x = x(t).$$

Sign convention

Choose positive directions first; a negative answer means change opposite the chosen positive direction.

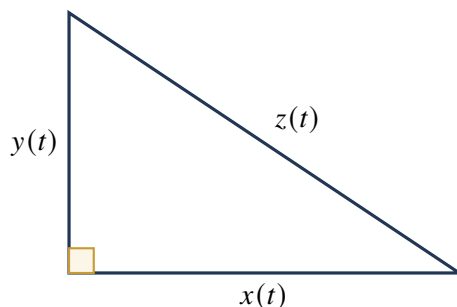
Similarity

In cone and shadow problems, use similar triangles to eliminate an extra variable before differentiating.

Concept Development and Strategy

1. Draw and label a diagram; identify constants, variables, known rates, and the requested rate.
2. Write one constraint containing the target variable and the variables whose rates are known.
3. Differentiate every changing quantity with respect to the same independent variable.
4. Only after differentiation, substitute the values at the specified instant and solve.
5. State the final rate with sign, units, and a complete contextual interpretation.

Shared Representation / Data



$$x^2 + y^2 = z^2$$

Differentiate with respect to t
before substituting an instant.

Worked Example: Expanding circle

A circle's radius increases at 3 cm/s. How fast is its area increasing when $r = 5$ cm?

Answer and Reasoning

$$A = \pi r^2, \text{ so } A' = 2\pi r r' = 2\pi(5)(3) = 30\pi \text{ cm}^2/\text{s}.$$

Worked Example: Ladder

A 10-m ladder rests against a wall. The base moves away at 0.6 m/s. Find the top's rate when the base is 6 m from the wall.

Answer and Reasoning

$$x^2 + y^2 = 100. \text{ At } x = 6, y = 8. \text{ Then } 2xx' + 2yy' = 0, \text{ so } y' = -\frac{6(0.6)}{8} = -0.45 \text{ m/s}.$$

Worked Example: Conical tank

Water in a cone satisfies $r/h = 1/3$. If $h' = 0.2$ m/min, find V' when $h = 6$ m.

Answer and Reasoning

$$r = h/3, \text{ so } V = \frac{1}{3}\pi(h^2/9)h = \frac{\pi}{27}h^3. \text{ Thus } V' = \frac{\pi}{9}h^2h' = \frac{\pi}{9}(36)(0.2) = 0.8\pi \text{ m}^3/\text{min}.$$

Worked Example: Approaching vehicles

One vehicle is 3 km east of an intersection moving east at 40 km/h; another is 4 km north moving north at 30 km/h. Find the rate of change of their separation.

Answer and Reasoning

$$z^2 = x^2 + y^2. \text{ } z = 5, \text{ so } z' = (3 \cdot 40 + 4 \cdot 30)/5 = 48 \text{ km/h; separation is increasing.}$$

Guided Practice and Discussion

Directions

Show the setup, preserve exact values unless a decimal is requested, and include units and theorem conditions whenever the context requires them.

1. Differentiate $x^2 + y^2 = 25$ with respect to t .

2. Differentiate $V = \frac{4}{3}\pi r^3$ with respect to time.

3. A circle area increases at $12\pi \text{ cm}^2/\text{s}$. Find r' when $r = 3 \text{ cm}$.

4. A cube volume increases at $54 \text{ cm}^3/\text{s}$. Find the side rate when $s = 3 \text{ cm}$.

5. Air enters a sphere at $100 \text{ cm}^3/\text{s}$. Find r' when $r = 4 \text{ cm}$.

6. A cone has fixed height 12 cm and changing radius. If $V' = 20\pi \text{ cm}^3/\text{s}$ when $r = 5$, find r' .

7. A right circular cone preserves volume while its height increases. Derive the relation between r'/r and h'/h .

AP Reasoning Check

Multiple-Choice Check

Choose the best answer, then identify the mathematical evidence supporting your choice.

1. In a related-rates problem, numerical values should usually be substituted

- | | |
|---------------------------------|---|
| (A) before writing a constraint | (B) before differentiating |
| (C) after differentiating | (D) only after solving the entire problem |

2. If $A = \pi r^2$, then

- | | |
|----------------------|-----------------------|
| (A) $A' = 2\pi r$ | (B) $A' = 2\pi r r'$ |
| (C) $A' = \pi(r')^2$ | (D) $A' = \pi r^2 r'$ |

3. A sphere's volume increases at $75 \text{ cm}^3/\text{s}$. At radius 3.2 cm, r' is closest to

- | | |
|----------|----------|
| (A) 0.29 | (B) 0.58 |
| (C) 1.83 | (D) 7.32 |

Common Errors and AP Precision

- State the quantity, sign, input value, and units in every contextual interpretation.
- In related rates, differentiate before substituting values. In linearization, identify the base point. For L'Hopital's Rule, verify $0/0$ or ∞/∞ first.
- A complete AP response names the definition, sign change, theorem, or comparison that supports the conclusion.

Exit Ticket

A particle on $x^2 + 4y^2 = 100$ has $x' = 5$ at $(6, 4)$. Find y' and interpret its sign.

Original instructional resource prepared for Education City High School. Not an official College Board publication.